



2219 - Shining light on the CO-dark H₂ gas in the heart of M83

Cycle: 1, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
M-83-MIRI-IFU_short				
	1	M-83-F1	MIRI Medium Resolution Spectroscopy	(2) M-83-F1
	5	M-83-Background-short	MIRI Medium Resolution Spectroscopy	(6) M-83-BACKGROUND
M-83-MIRI-IFU_medium				
	6	M-83-F1	MIRI Medium Resolution Spectroscopy	(2) M-83-F1
	10	M-83-F1-F2-Background	MIRI Medium Resolution Spectroscopy	(6) M-83-BACKGROUND
M-83-MIRI-IFU_long				
	11	M-83-F1	MIRI Medium Resolution Spectroscopy	(2) M-83-F1

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
	15	M-83-F1-F2-Background	MIRI Medium Resolution Spectroscopy	(6) M-83-BACKGROUND

ABSTRACT

Molecular gas is a critical ingredient in the recipe of star formation (SF) in galaxies. In order to fully understand the processes that govern SF, it is essential to accurately measure, map and characterize the distribution of H₂ in star-forming environments. Since H₂ is a weak rotational emitter, the molecular gas content in galaxies is typically inferred using indirect tracers. The CO (1-0) transition has been widely used as H₂ tracer for decades. However, CO provides a partial census of the total H₂ mass, particularly in regions with large quantities of CO-dark gas. Recent evidence from an FUV spectroscopic study suggests that S⁺ might be tracing large amounts of CO-dark gas in the core of M83. Here we propose to exploit the unprecedented capabilities of JWST in the MIR, using the MIRI/MRS, to perform a spatially-resolved study of the warm H₂ gas in the heart of this face-on spiral galaxy. Our motivation is to directly detect the warm H₂ in the core of M83, and infer the total H₂ mass by combining the spatially-resolved data with a new continuous-temperature model using the H₂ rotational emission. Ultimately, we will be able to test if S⁺ does indeed trace CO-dark gas in such environments with intense SF. To understand the fueling SF history through cosmic time, it is imperative that we test and develop tools to accurately estimate molecular gas mass directly probing the H₂ reservoirs. JWST, with its unrivaled sensitivity will allow us to do exactly that.

OBSERVING DESCRIPTION

We will observe the nucleus of the nearby spiral galaxy, M83, with the MIRI/MRS IFU. We will require four pointings to fully cover the nucleus of M83 with Channel 4 (6.9" X 7.9"), and fully sample the massive stellar clusters in all four channels. In addition, we request companion background observations with identical integrations as that of the individual M83 pointings to accurately measure and correct for the thermal background.

Given the expected absolute pointing accuracy of JWST (0.10"), we do not request additional time to perform a target acquisition. Our proposed science goals can be achieved with a pointing uncertainty of ~0.10".

We will obtain MIRI/MRS IFU spectral maps to accurately trace the warm molecular gas in the core of this galaxy. To fully exploit the potential of MIRI/MRS we will cover the full wavelength range, ~5 - 28 micron using all three grating settings, SHORT (A), MEDIUM (B), and LONG (C), and the FAST readout mode. We will collect simultaneous off-source imaging observations using the F1000W, F1130W, and F2100W filters. With these observations we will primarily map PAH and warm dust emission arising on the disk of M83.

JWST Proposal 2219 (Created: Tuesday, May 31, 2022 at 12:01:18 PM Eastern Standard Time) - Overview

We do not require any specific PA orientation for the MIRI/MRS pointings in the nucleus, however, we constrain the PA range between 275 and 15° for the MIRI/MRS background observations. This constraint is particularly relevant for the MIRI imaging of the background pointing, as we need to avoid the disk of the galaxy.

Proposal 2219 - Targets - Shining light on the CO-dark H2 gas in the heart of M83

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	M-83-F1	RA: 13 37 0.7100 (204.2529583d) Dec: -29 51 58.60 (-29.86628d) Equinox: J2000		
	Comments: This point was extracted from the WFC3 IF images from PID:11360 Category=Galaxy Description=[Starburst galaxies]				
	(6)	M-83-BACKGROUND	RA: 13 37 30.2059 (204.3758579d) Dec: -29 56 1.36 (-29.93371d) Equinox: J2000		
	Comments: Category=Calibration Description=[Telescope/sky background]				

Proposal 2219 - Observation 1 - Shining light on the CO-dark H2 gas in the heart of M83

Observation	Proposal 2219, Observation 1: M-83-F1												Tue May 31 17:01:19 GMT 2022
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[M-83-Background-short (Obs 5)]												
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(2)	M-83-F1	RA: 13 37 0.7100 (204.2529583d) Dec: -29 51 58.60 (-29.86628d) Equinox: J2000										
	Comments: This point was extracted from the WFC3 IF images from PID:11360												
	Category=Galaxy Description=[Starburst galaxies]												
Acquisition	#							Target					
	1							NONE					
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
		ALL				YES				FULL			
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift		Column shift		Tile Order		
	2	2	-50.0		-74.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1000W	FASTR1	40	5	1	Dither 1	4	20	2264.433	61359
	1	SHORT(A)	MRSLONG		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162
	1	SHORT(A)	MRSSHORT		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162

Proposal 2219 - Observation 1 - Shining light on the CO-dark H2 gas in the heart of M83

Special Requirements	Sequence Observations 1, 5, Non-interruptible
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Proposal 2219 - Observation 5 - Shining light on the CO-dark H2 gas in the heart of M83

Observation	Proposal 2219, Observation 5: M-83-Background-short												Tue May 31 17:01:19 GMT 2022	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [M-83-F1 (Obs 1)]													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	M-83-BACKGROUND			RA: 13 37 30.2059 (204.3758579d) Dec: -29 56 1.36 (-29.93371d) Equinox: J2000									
	Comments: Category=Calibration Description=[Telescope/sky background]													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				
		ALL				YES				FULL				
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1000W	FASTR1	40	5	1	Dither 1	4	20	2264.433	61359	
	1	SHORT(A)	MRSLONG		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	
	1	SHORT(A)	MRSSHORT		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	

Proposal 2219 - Observation 5 - Shining light on the CO-dark H2 gas in the heart of M83

Special Requirements	Sequence Observations 1, 5, Non-interruptible
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Proposal 2219 - Observation 6 - Shining light on the CO-dark H2 gas in the heart of M83

Observation	Proposal 2219, Observation 6: M-83-F1												Tue May 31 17:01:19 GMT 2022	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[M-83-F1-F2-Background (Obs 10)]													
Diagnostics	(Visit 6:1) Warning (Form): Data Excess over lower threshold													
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous			
	(2)	M-83-F1		RA: 13 37 0.7100 (204.2529583d)										
				Dec: -29 51 58.60 (-29.86628d)										
				Equinox: J2000										
	Comments: This point was extracted from the WFC3 IF images from PID:11360													
	Category=Galaxy													
	Description=[Starburst galaxies]													
Acquisition	#											Target		
	1											NONE		
Template	AcqFilter	Primary Channel			Simultaneous Imaging				Imager Subarray					
		ALL			YES				FULL					
Mosaic	Rows	Columns		Row Overlap %		Column Overlap %		Row shift		Column shift		Tile Order		
	2	2		-50.0		-74.0		0.0		0.0		DEFAULT		
Dithers	#	Dither Type			Optimized For				Direction					
	1	4-Point			EXTENDED SOURCE				NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1130W	FASTR1	40	5	1	Dither 1	4	20	2264.433	61359	
	1	MEDIUM(B)	MRSLONG		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	
	1	MEDIUM(B)	MRSSHORT		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	

Proposal 2219 - Observation 6 - Shining light on the CO-dark H2 gas in the heart of M83

Special Requirements	Sequence Observations 6, 10, Non-interruptible
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Proposal 2219 - Observation 10 - Shining light on the CO-dark H2 gas in the heart of M83

Observation	Proposal 2219, Observation 10: M-83-F1-F2-Background												Tue May 31 17:01:19 GMT 2022
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [M-83-F1 (Obs 6)]												
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(6)	M-83-BACKGROUND	RA: 13 37 30.2059 (204.3758579d) Dec: -29 56 1.36 (-29.93371d) Equinox: J2000										
	Comments: Category=Calibration Description=[Telescope/sky background]												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray			
		ALL				YES				FULL			
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1130W	FASTR1	40	5	1	Dither 1	4	20	2264.433	61359
	1	MEDIUM(B)	MRSLONG		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162
	1	MEDIUM(B)	MRSSHORT		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162

Proposal 2219 - Observation 10 - Shining light on the CO-dark H2 gas in the heart of M83

Special Requirements	Sequence Observations 6, 10, Non-interruptible
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Proposal 2219 - Observation 11 - Shining light on the CO-dark H2 gas in the heart of M83

Observation	Proposal 2219, Observation 11: M-83-F1												Tue May 31 17:01:19 GMT 2022	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[M-83-F1-F2-Background (Obs 15)]													
Diagnostics	(M-83-F1 (Obs 11)) Warning (Form): Imager Filter overlap.													
	(Visit 11:1) Warning (Form): Data Excess over lower threshold													
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	M-83-F1	RA: 13 37 0.7100 (204.2529583d) Dec: -29 51 58.60 (-29.86628d) Equinox: J2000											
	Comments: This point was extracted from the WFC3 IF images from PID:11360													
	Category=Galaxy Description=[Starburst galaxies]													
Acquisition	#											Target		
	1											NONE		
Template	AcqFilter	Primary Channel				Simultaneous Imaging				Imager Subarray				
		ALL				YES				FULL				
Mosaic	Rows	Columns	Row Overlap %		Column Overlap %		Row shift		Column shift		Tile Order			
	2	2	-50.0		-74.0		0.0		0.0		DEFAULT			
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/E xp	Exposures/Dit h	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F2100W	FASTR1	40	5	1	Dither 1	4	20	2264.433	61359	
	1	LONG(C)	MRSLONG		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	
	1	LONG(C)	MRSSHORT		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	

Proposal 2219 - Observation 11 - Shining light on the CO-dark H2 gas in the heart of M83

Special Requirements	Sequence Observations 11, 15, Non-interruptible
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Proposal 2219 - Observation 15 - Shining light on the CO-dark H2 gas in the heart of M83

Observation	Proposal 2219, Observation 15: M-83-F1-F2-Background												Tue May 31 17:01:19 GMT 2022	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [M-83-F1 (Obs 11)]													
Diagnostics	(M-83-F1-F2-Background (Obs 15)) Warning (Form): Imager Filter overlap.													
	(Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections				Miscellaneous			
	(6)	M-83-BACKGROUND		RA: 13 37 30.2059 (204.3758579d)										
				Dec: -29 56 1.36 (-29.93371d)										
				Equinox: J2000										
Acquisition	Comments:													
	Category=Calibration													
	Description=[Telescope/sky background]													
	#	Target												
	1	NONE												
Template	AcqFilter		Primary Channel			Simultaneous Imaging				Imager Subarray				
			ALL			YES				FULL				
Dithers	#	Dither Type			Optimized For				Direction					
	1	4-Point			EXTENDED SOURCE				NEGATIVE					
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F2100W	FASTR1	40	5	1	Dither 1	4	20	2264.433	61359	
	1	LONG(C)	MRSLONG		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	
	1	LONG(C)	MRSSHORT		FASTR1	40	5	1	Dither 1	4	20	2264.433	59162	

Proposal 2219 - Observation 15 - Shining light on the CO-dark H2 gas in the heart of M83

Special Requirements	Sequence Observations 11, 15, Non-interruptible
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